

Algebra II with Trigonometry

Chapter 5.5

Mental Math (GPT-5.4)

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- 1. Law of Sines Match-Up** In triangle ABC , side a is opposite angle A . Which angle should be paired with side a in the Law of Sines?
- 2. Third Angle First** A triangle has angles 50° and 68° . What is the third angle?
- 3. Equal Sides, Equal Angles** If two sides of a triangle are equal, what must be true about their opposite angles?
- 4. Bigger Side or Bigger Angle?** In a triangle, side c is the longest side. What can you conclude about angle C ?
- 5. Acute SSA: Two, One, or None?** For an SSA triangle with acute angle A , if $b > a > b \sin A$, how many triangles are possible?
- 6. Acute SSA Boundary Case** For an SSA triangle with acute angle A , if $a = b \sin A$, how many triangles are possible?
- 7. Acute SSA Impossible Case** For an SSA triangle with acute angle A , if $a < b \sin A$, how many triangles are possible?
- 8. Obtuse SSA Check** For an SSA triangle with obtuse angle A , if $a \leq b$, is there a triangle?
- 9. Supplement Test** If the Law of Sines gives $\sin B = \sin 40^\circ$, what is the other possible angle value for B before checking whether it makes a valid triangle?

10. **Height in the Ambiguous Case** In the SSA ambiguous case, what formula is used to find the height h of the triangle first?